

$$\underline{LS(a)} = A_4^{tree} \quad | \quad \underline{\text{Leading Singularity}}$$

$$\underline{\text{Proof}}: LS(a) =$$

$$\int \left[\prod_{i=1}^4 d^4 p_i \right] (A_2^{\overline{tree}}(-l_1, 1, l_1) A_3^{\overline{tree}}(-l_1, p_2, l_2) A_3^{\overline{tree}}(-l_2, p_3, l_3) A_3^{\overline{tree}}(-l_2, p_4, l_4)) |_{l=l(a)}$$

$$\textcircled{A} \left(\partial_{l_4}^{(u)} \delta^{o1u}(-l_1, 1, l_1) \delta^{o18}(-l_2, p_4, l_4) + \delta^{o1u}(-l_1, l_1) \partial_{l_4}^{(u)} \delta^{o18} \right) \delta^{o1u}(-l_2, p_3, l_3)$$

$$\textcircled{B} \left((\partial_{l_3}^{(u)} \delta^{o18}) (\partial_{l_4} \delta^{o1u}(l_4, 1, l_1)) + \delta^{o1u}(-l_1, 1, l_1) \partial_{l_3} \partial_{l_4} \delta^{o18} \right) \delta^{o1u}(-l_2, p_3, l_3) \\ + (\partial_{l_4} \delta^{o1u}(-l_1, 1, l_1) \delta^{o18}(-l_2, p_4, l_4) + \delta^{o1u}(-l_1, 1, l_1) \partial_{l_4} \delta^{o18}) \partial_{l_3}^{(u)} \delta^{o1u}(-l_2, p_3, l_3)$$

$$\textcircled{C} \left[\partial_{l_2}^{(u)} (\delta^{o1u}(-l_2, p_3, l_3)) \right] \left[\partial_{l_3}^{(u)} \delta^{o18} \partial_{l_4} \delta^{o1u}(l_4, 1, l_1) + \delta^{o1u}(-l_1, 1, l_1) \partial_{l_3} \partial_{l_4} \delta^{o18} \right]$$

$$\textcircled{D} \underbrace{\partial_{l_2}^{(u)} \delta^{o1u}(-l_2, p_3, l_3)}_{[3l_3]^4} \underbrace{\left[\partial_{l_1}^{(u)} \delta^{o1u}(-l_1, 1, l_1) \partial_{l_3}^{(u)} \partial_{l_4}^{(u)} \delta^{o18}(-l_2, p_4, l_4) \right]}_{[l_4]^4 \angle l_3 l_4^4}$$

$$= \int [3l_3]^4 \delta^{o18}(\hat{\phi}) [l_4]^4 \angle l_3 l_4^4$$

$$\frac{[l_1][l_1 l_4] \angle l_1 2 \angle 2 l_2 \angle l_2 l_1 [l_2 3] [3l_3] [l_3 l_2] \angle l_3 4 \angle 4 l_4 \angle l_4 l_3}{[l_1][l_1 l_4] \angle l_1 2 \angle 2 l_2 \angle l_2 l_1 [l_2 3] [3l_3] [l_3 l_2] \angle l_3 4 \angle 4 l_4 \angle l_4 l_3}$$

$$= \int [3l_3]^3 \delta^{o18}(\hat{\phi}) [l_4]^3 \angle l_3 l_4^3$$

$$\frac{[l_1][l_1 l_4] \angle l_1 2 \angle 2 l_2 \angle l_2 l_1 [l_2 3] [3l_3] [l_3 l_2] \angle l_3 4 \angle 4 l_4 \angle l_4 l_3}{[l_1][l_1 l_4] \angle l_1 2 \angle 2 l_2 \angle l_2 l_1 [l_2 3] [3l_3] [l_3 l_2] \angle l_3 4 \angle 4 l_4 \angle l_4 l_3}$$

$$\textcircled{E} [3l_3] \angle l_3 l_4 [l_4]$$

$$= [3][12] \angle l_1 + [13] \angle 3 + [14] \angle 4) (l_1 \angle [l_1 + l_4] [0]) [1]$$

$$= [3] ([12] \angle l_1 + [14] \angle 4) ([1] \angle [l_1 + l_4] [0]) [1]$$

$$= [34] \angle l_4 \angle [l_1] + [32] \angle l_1 \angle [4]$$

$$= \angle l_4 ([32] [4] - [34] [2])$$

$$\left[\begin{array}{l} [32] [4] - [34] [2] = -[31] [24] \\ \rightarrow -\angle l_4 [31] [24] \end{array} \right]$$

$$= \frac{[13]^3 \angle l_4^2 [l_4]^3 \int \delta^{o18}(\hat{\phi})}{[13]^3 \angle l_4^2 [l_4]^3 \int \delta^{o18}(\hat{\phi})}$$

$$[l_1][l_1 l_4] \angle l_1 2 \angle l_2 \angle l_2 l_1 [l_2 3] [3l_3] [l_3 l_2] \angle l_3 4 \angle l_4 l_3$$

$$\angle l_1 l_2 [l_2 3] = -\angle l_1 2 [13]$$

$$[13]^2 [l_4]^3 [l_4]^3$$

$$\langle l_1 \rangle \langle l_2 \rangle \langle l_3 \rangle \langle l_4 \rangle \langle l_1 l_2 \rangle \langle l_2 l_3 \rangle \langle l_3 l_4 \rangle \langle l_4 l_1 \rangle$$

Claim:
$$\int \prod_{i=1}^4 d^4 x_i A_3^{\mu_1 \mu_2 \mu_3}(-l_2, p_1, l_1) A_3^{\mu_4 \mu_5 \mu_6}(-l_1, p_2, l_2) A_3^{\mu_7 \mu_8 \mu_9}(-l_2, p_3, l_3) A_3^{\mu_{10} \mu_{11} \mu_{12}}(-l_3, p_4, l_4) \Big|_{l=l^{(4)}}$$

where
$$\mathcal{J} \equiv \left(\frac{1}{y_{12}^2 y_{34}^2} \right) = \frac{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle}{\langle 1234 \rangle^2} (-1)$$

*
$$y_i^\mu = x_i^\mu - x_{i+1}^\mu = p_i^\mu$$

$$y_{ij}^\mu = y_i^\mu - y_j^\mu = p_i^\mu + \dots + p_j^\mu$$

$$\langle 1234 \rangle = -\epsilon_{IJKL} Z_1^I Z_2^J Z_3^K Z_4^L$$

*
$$A_3^{\mu_1 \mu_2 \mu_3}(-l_2, p_1, l_1) A_3^{\mu_4 \mu_5 \mu_6}(-l_1, p_2, l_2)$$

$$\Rightarrow A_3^{\mu_1 \mu_2 \mu_3}(-l_2, p_1, l_1) \quad + + -$$

①
$$A_3^{\mu_1 \mu_2 \mu_3}(-l_2, p_1, l_1) = \langle l_2 p_1 \rangle^c \langle l_2 l_1 \rangle^b \langle p_1 l_1 \rangle^a$$

